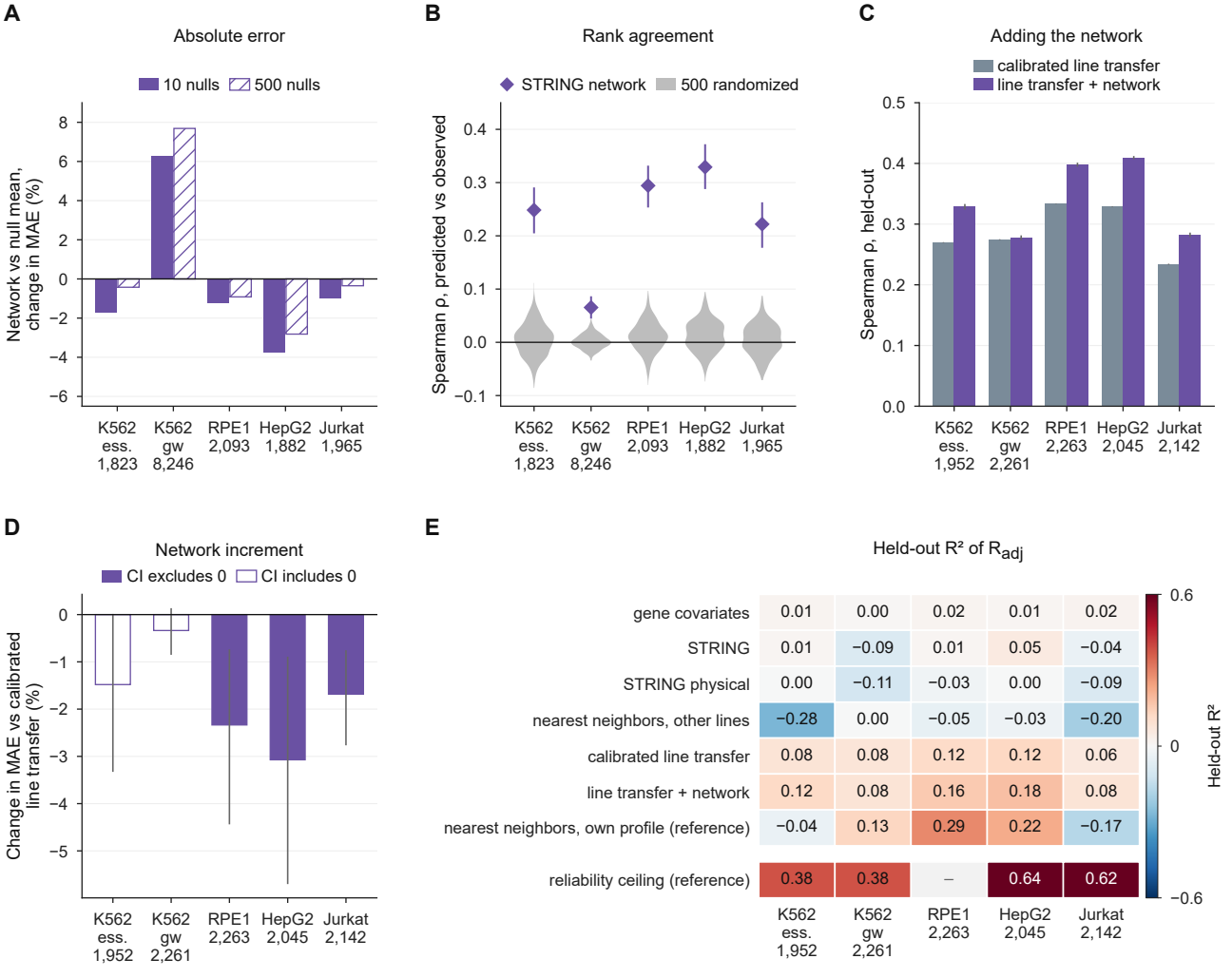

A
Absolute error
10 nulls
500 nulls
Network vs null mean, change in MAE (%)
8
6
4
2
0
−2
−4
−6
K562 ess. 1,823
K562 gw 8,246
RPE1 2,093
HepG2 1,882
Jurkat 1,965
B
Rank agreement
STRING network
500 randomized
Spearman ρ, predicted vs observed
0.4
0.3
0.2
0.1
0.0
−0.1
K562 ess. 1,823
K562 gw 8,246
RPE1 2,093
HepG2 1,882
Jurkat 1,965
C
Adding the network
calibrated line transfer
line transfer + network
Spearman ρ, held-out
0.5
0.4
0.3
0.2
0.1
0.0
K562 ess. 1,952
K562 gw 2,261
RPE1 2,263
HepG2 2,045
Jurkat 2,142
D
Network increment
CI excludes 0
CI includes 0
Change in MAE vs calibrated line transfer (%)
0
−1
−2
−3
−4
−5
K562 ess. 1,952
K562 gw 2,261
RPE1 2,263
HepG2 2,045
Jurkat 2,142
E
Held-out R² of R_adj
gene covariates 0.01 0.00 0.02 0.01 0.02
STRING 0.01 −0.09 0.01 0.05 −0.04
STRING physical 0.00 −0.11 −0.03 0.00 −0.09
nearest neighbors, other lines −0.28 0.00 −0.05 −0.03 −0.20
calibrated line transfer 0.08 0.08 0.12 0.12 0.06
line transfer + network 0.12 0.08 0.16 0.18 0.08
nearest neighbors, own profile (reference) −0.04 0.13 0.29 0.22 −0.17
reliability ceiling (reference) 0.38 0.38 – 0.64 0.62
K562 ess. 1,952
K562 gw 2,261
RPE1 2,263
HepG2 2,045
Jurkat 2,142
0.6
0
−0.6
Held-out R²